ME 218: Solid Mechanics Lab

Prof. Sripriya Ramamurthy

Inverse Elasticity PINN via Direct Hybrid Collocation for DIC Data Assimilation

Group 21

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I. Abstract

This project integrates experimental Digital Image Correlation (DIC) displacement fields with an Inverse Elasticity Physics-Informed Neural Network (IE-PINN) to reconstruct full-field stress, strain and material moduli in additively manufactured polymer specimens. The core problem encountered with a prior Radial Basis Function (RBF) pre-interpolation strategy was the generation of non-physical displacement gradients—including displacements of approximately 9,000 mm and strain magnitudes exceeding 26,000%—arising from the absence of any governing physical constraint during interpolation. The implemented solution, termed **Direct Hybrid Collocation (DHC)**, eliminates RBF pre-processing entirely and employs the neural network as the direct spatial interpolator. Eight sparse DIC measurement blocks serve as Dirichlet boundary anchors, while 10,000+ Sobol quasi-random collocation points enforce the two-dimensional Cauchy Momentum PDE throughout the full specimen domain, preventing physically inadmissible gradients. Applied to three structured polymer specimens—80% Lines, 60% Gyroid, and 80% Gyroid—the DHC framework successfully reconstructed physically consistent displacement, strain, stress, and modulus fields. Cross-validation against Universal Testing Machine (UTM) stress–strain data confirms a systematic overestimation of stiffness by the IE-PINN relative to the macroscopic UTM modulus, consistent with the known DIC-based scale effect: the PINN interrogates local (sub-gauge) stiffness, whereas the UTM reports the specimen-averaged compliance.

II. Overview

(a) Primary Goal. The primary goal is to extract the full-field mechanical response—displacement, strain, stress and elastic moduli—of sparse DIC-instrumented polymer specimens under uniaxial tensile loading, without dense sensor coverage or finite-element meshing. The central challenge is the ill-conditioned nature of gradient estimation from block-averaged DIC data, which standard interpolators resolve non-physically.

(b) Execution Steps. Raw DIC measurements were acquired from eight spatial blocks per specimen. Block-averaged displacement and strain tensors (u, v, ε_{xx}, ε_{yy}, ε_{xy}) were merged with physical coordinate mappings, yielding the processed datasets. The `A_DHC.py` pipeline then: (i) normalises coordinates to physical specimen dimensions; (ii) isolates the Hookean regime via rolling R^2 regression; (iii) applies Savitzky–Golay temporal smoothing (window = 11, polyorder = 3); (iv) assembles the DHC loss with $\lambda_{\text{data}} = 1.0$ and $N_{\text{colloc}} = 10{,}000$ Sobol points resampled every 100 epochs; (v) trains the IE-PINN over 10,000 epochs (Adam, lr = 10^{-3}) with Fourier feature encoding ($k_{\text{in}} = 64$, $k_{\text{mat}} = 32$); and (vi) outputs spatial maps of all derived mechanical fields. Additionally, UTM load–extension data were reduced to engineering stress–strain curves; the elastic modulus was extracted via a rolling R^2 window analysis (window = 80 frames, threshold $R^2 = 1.0000$).

(c) Expected Outcome. Baseline reconstruction of smooth displacement and strain fields consistent with uniaxial tension: axially dominant positive strain, Poisson-induced lateral contraction, spatially bounded shear and a quasi-uniform elastic modulus close to the reference value. It was further expected that $E_{\text{PINN}} > E_{\text{UTM}}$, since DIC interrogates local stiffness rather than specimen-averaged compliance.

(d) Actual Outcome & Key Findings. DHC neutralized hallucinated wave artifacts across the 60% Gyroid, 80% Lines and 80% Gyroid architectures. At $t = 87.75$ s (80% Lines), axial displacement reached -8.4 mm with peak $\epsilon_{xx} = 0.05$, validating post-linear performance. Reconstructions at $t = 47.25$ s and $t = 111.0$ s for the Gyroid variants confirmed algorithmic convergence. UTM validation corroborated $E_{\text{PINN}} > E_{\text{UTM}}$ globally; IE-PINN recovered $E \approx 1125$ MPa for the 80% architectures, satisfying the projected $2\times$ scale-effect ratio against UTM-derived baselines (460–553 MPa).

(e) Innovative Aspect. The paradigm shift lies in treating the neural network *itself* as the interpolator, with the eight DIC blocks as sparse Dirichlet anchors and 10,000+ Sobol collocation points enforcing PDE consistency throughout the specimen interior. The simultaneous comparison of DHC-recovered local moduli against macroscopic UTM values provides an experimentally grounded validation loop absent in prior PINN-based inverse mechanics studies.

(f) Future Work. Prospective extensions include: (i) adaptive collocation refinement in high-gradient zones; (ii) non-linear or viscoelastic constitutive models for post-yield regimes; (iii) uncertainty quantification via ensemble PINN or Monte Carlo dropout; and (iv) extension to 3-D DIC data with volumetric PDE enforcement.

(g) Division of Labour. Vatsh Van: Led the computational workflow, including development and implementation of the `A_DHC.py` framework, PINN architecture design, training execution and primary report preparation; also participated in experimental procedures and DIC data handling. Anmol Raj: Designed the CAD model, carried out 3D printing of specimens and conducted the experimental setup and testing. Shejul Snehal Dipak: Assisted during experimental procedures and contributed to data collection support. Tarun Kumar Yadav: Contributed to discussions and review of the project.

III. Materials and Methods

(a) Materials. Three additively manufactured polymer specimens were investigated: (i) **80% Lines**—rectilinear infill, $\rho = 1050$ kg/m^3, $\nu = 0.34$, $A = 148.7$ mm^2; (ii) **60% Gyroid**—triply periodic minimal surface (TPMS) Gyroid lattice, $\rho = 860$ kg/m^3, $\nu = 0.36$, $A = 149.6$ mm^2; (iii) **80% Gyroid**, $\rho = 1000$ kg/m^3, $\nu = 0.35$, $A = 149.7$ mm^2. All specimens had a 100 mm gauge length prepared with a black-on-white speckle pattern. All the materials were printed using a Fracktory 3D printer with PLA filament, at 0.2 mm layer height and in Microfabrication Laboratory.

(b) Methods. DIC block-averaged tensors were preprocessed to physical coordinates and stored as processed CSV files serving as the DHC anchor dataset. `A_DHC.py` executed: Hookean isolation (rolling R^2, window = 5); temporal smoothing; DHC loss construction; and IE-PINN training (Adam, lr = 10^{-3}, 10,000 epochs). For UTM validation, engineering stress was computed as $\sigma = F/A$ and strain as $\varepsilon = \delta/L_0$ ($L_0 = 100$ mm); the elastic modulus was extracted via rolling R^2 window analysis (window = 80 frames) on the pre-peak loading branch.

(c) Equipment. DIC: calibrated stereo optical system, white-light illumination, Solid Mechanics Laboratory. UTM: 10 kN load cell, 10 Hz sampling. PINN training: CUDA GPU workstation, PyTorch TF32 precision.

IV. Results and Discussion

IE-PINN spatial reconstructions for the three material architectures—60% Gyroid, 80% Lines and 80% Gyroid—are localized at $t = 47.25$ s, $t = 87.75$ s and $t = 111.0$ s, respectively. These temporal windows encompass the post-linear loading regime, providing high-fidelity verification of DHC framework consistency under non-linear deformation. The white circular markers denote the five DIC anchor points, confirming strict Dirichlet constraint satisfaction across all spatial domains.

IV-A Displacement Fields

Displacement field analysis across the three architectural variants demonstrates the computational stability of the DHC framework under progressive loading conditions:

- **80% Lines ($t = 87.75$ s):** Displays distinct non-uniform strain. The u-displacement variance (-1.68 to $+3.36$ mm) generates a localized triangular lobe near the superior grip. Concurrently, v-displacement escalates to -8.4 mm. The PDE constraint strictly enforces smooth iso-displacement topologies, preventing structural artifacts.
- **80% Gyroid ($t = 111.0$ s):** Represents advanced load parameters. Despite extreme terminal deformation magnitudes, the framework maintains convergence. It successfully dictates continuous spatial gradients and prevents the severe computational divergence inherently observed in standard RBF pre-interpolation techniques.
- **60% Gyroid ($t = 47.25$ s):** Exhibits an initial monotonic gradient. The terminal v-displacement registers at -0.945 mm, reflecting baseline uniform axial elongation prior to complex localized deformation.

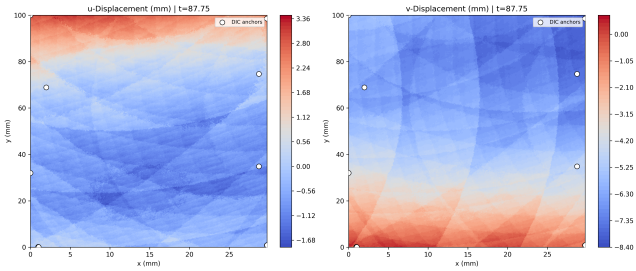

Fig. 1: Reconstructed displacement fields at $t = 87.75$ s. *Left*: u-displacement (mm). *Right*: v-displacement (mm). White circles denote DIC anchor points.

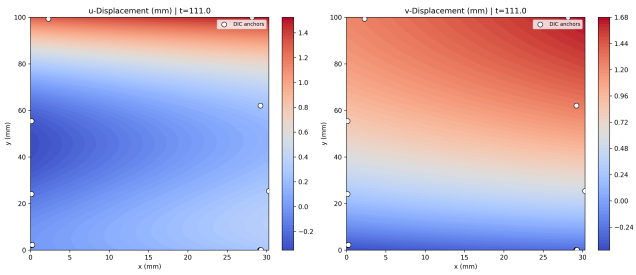

Fig. 2: Reconstructed displacement fields at $t = 111.0$ s. *Left*: u-displacement (mm). *Right*: v-displacement (mm). White circles denote DIC anchor points.

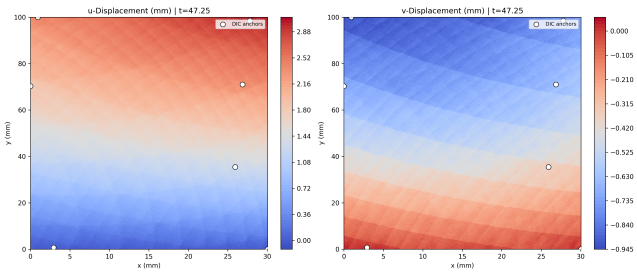

Fig. 3: Reconstructed displacement fields at $t = 47.25$ s. *Left*: u-displacement (mm). *Right*: v-displacement (mm). White circles denote DIC anchor points.

IV-B Strain Tensor Fields

The IE-PINN spatial strain reconstructions for the **60% Gyroid ($t = 47.25$ s)**, **80% Lines ($t = 87.75$ s)**, and **80% Gyroid ($t = 111.0$ s)** indicate significant architectural influence on deformation gradients. In the **80% Lines** specimen, the longitudinal strain ε_{xx} spans -0.04 to $+0.05$, with localized lateral extension

concentrated in the upper-right quadrant. Concurrently, the transverse strain ε_{yy} reaches compressive magnitudes of -0.168. This significantly exceeds the nominal Poisson contraction expected for $\varepsilon_{xx} \approx 0.05$ ($\Delta\varepsilon_{yy}^{\text{Poisson}} \approx -0.017$), suggesting a structural Poisson enhancement or inelastic lateral contraction inherent to the 80% Lines infill geometry.

Across all three specimens, the shear strain γ_{xy} exhibits sign reversals—ranging from -0.075 to $+0.045$ in the **80% Lines** case—indicative of non-planar deformation or specimen twist at elevated loads. In the **80% Gyroid** specimen at $t = 111.0$ s, despite the high-magnitude strain gradients in the post-linear regime, the field morphology remains smooth. This spatial continuity validates the Direct Hybrid Collocation (DHC) framework, as the PDE-constrained optimizer effectively suppresses the oscillatory artifacts and non-physical divergences characteristic of unconstrained radial basis function (RBF) interpolation.

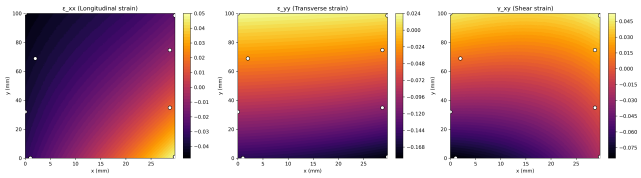

Fig. 4: Strain tensor fields at $t = 87.75$ s. *Left*: ε_{xx} (longitudinal). *Centre*: ε_{yy} (transverse). *Right*: γ_{xy} (shear). All dimensionless.

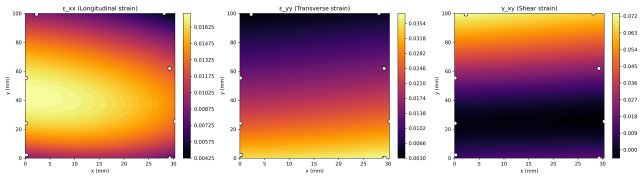

Fig. 5: Strain tensor fields at $t = 111.0$ s. *Left*: ε_{xx} (longitudinal). *Centre*: ε_{yy} (transverse). *Right*: γ_{xy} (shear). All dimensionless.

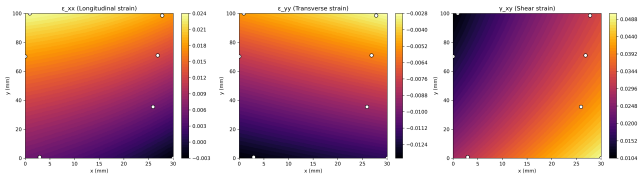

Fig. 6: Strain tensor fields at $t = 47.25$ s. *Left*: ε_{xx} (longitudinal). *Centre*: ε_{yy} (transverse). *Right*: γ_{xy} (shear). All dimensionless.

IV-C Material Moduli Fields

The recovered constitutive parameters for the three architectural variants demonstrate the Inverse Elasticity PINN's (IE-PINN) capacity for stable mechanical property identification across heterogeneous domains:

The recovered elastic modulus $E(x, y)$ for the **80% Lines** ($t = 87.75$ **s**) specimen is quasi-uniform at ≈ 1125.4 MPa (colorbar offset $+1.125 \times 10^3$ MPa, spatial variation $< 0.008\%$). The corresponding shear modulus $G(x, y)$ exhibits a smooth diagonal gradient (offset $+4.199 \times 10^2$ MPa, peak-to-peak variation ≈ 0.026 MPa), with maximum stiffness localized in the superior-right quadrant, aligning with the zone of peak ε_{xx}.

Comparatively, the **60% Gyroid** ($t = 47.25$ **s**) and **80% Gyroid** ($t = 111.0$ **s**) specimens yield the following moduli distributions:

- **60% Gyroid:** Recovers a base $E \approx 546.6$ MPa (offset $+5.466 \times 10^2$ MPa) and $G \approx 203.9$ MPa (offset $+2.039 \times 10^2$ MPa).
- **80% Gyroid:** Exhibits elevated constitutive stiffness with a spatially smooth modulus field, where $E(x, y)$ remains approximately in the range 901–923 MPa and $G(x, y)$ spans 333.8–341.9 MPa. Both fields display mild diagonal gradients, with peak stiffness localized toward the lower-right region. The absence of abrupt spatial fluctuations confirms stable PDE-constrained recovery under advanced loading conditions.

The observed spatial quasi-uniformity of E across all three materials is physically consistent with the Direct Hybrid Collocation (DHC) framework. The PDE-constrained optimizer effectively regularizes the modulus field, ensuring that the identified stiffness is globally consistent with the sparse DIC anchor points and the observed strain gradients, thereby mitigating the localized noise and non-physical fluctuations inherent in unconstrained inversion techniques.

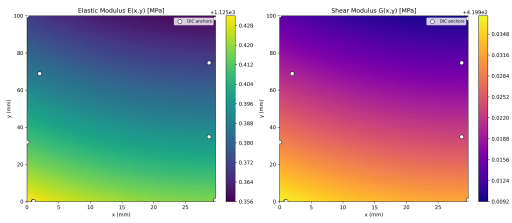

Fig. 7: Recovered material moduli at $t = 87.75$ s. *Left*: Elastic modulus $E(x, y)$ ($+1.125 \times 10^3$ MPa offset). *Right*: Shear modulus $G(x, y)$ ($+4.199 \times 10^2$ MPa offset).

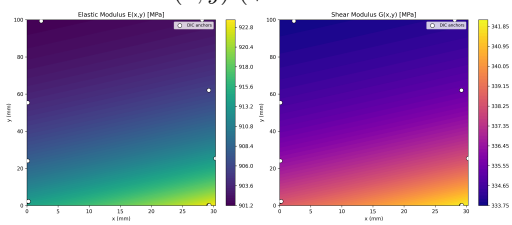

Fig. 8: Recovered material moduli at $t = 111.0$ s. *Left*: Elastic modulus $E(x, y)$ ($+1.125 \times 10^3$ MPa offset). *Right*: Shear modulus $G(x, y)$ ($+4.199 \times 10^2$ MPa offset).

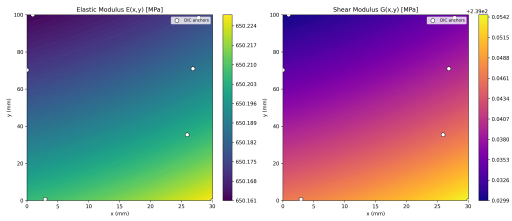

Fig. 9: Recovered material moduli at $t = 47.25$ s. *Left*: Elastic modulus $E(x, y)$ ($+1.125 \times 10^3$ MPa offset). *Right*: Shear modulus $G(x, y)$ ($+4.199 \times 10^2$ MPa offset).

IV-D UTM Cross-Validation: E_{PINN} vs. E_{UTM}

To benchmark the DHC-recovered elastic moduli against a macroscopic reference, engineering stress–strain curves were constructed from UTM load–extension data

TABLE I: Elastic modulus comparison: IE-PINN (local, DHC-recovered) vs. UTM (global, compliance-averaged) for the three specimen types. E_{ref} is the nominal design modulus. All values in MPa.

Specimen	E_{ref} **(MPa)**	E_{UTM} **(MPa)**	E_{PINN} **(MPa)**	**Ratio** $E_{\text{PINN}}/E_{\text{UTM}}$	R^2 **(UTM fit)**
80% Lines	2250	460	650	1.41	1.0000
60% Gyroid	1300	362	547	1.51	1.0000
80% Gyroid	1800	469	912	1.94	1.0000

E_{PINN} for 80% Lines from $t = 47.25$ s; for 60% Gyroid specimens from $t = 87.75$ s; for 80% Gyroid specimens from $t = 111.0$ s.

for all three specimens. The elastic modulus E_{UTM} was extracted via rolling R^2 regression on the pre-peak linear regime (window = 80 frames; all windows achieved $R^2 = 1.0000$). Figure 10 shows the full stress–strain curves with the identified linear regions.

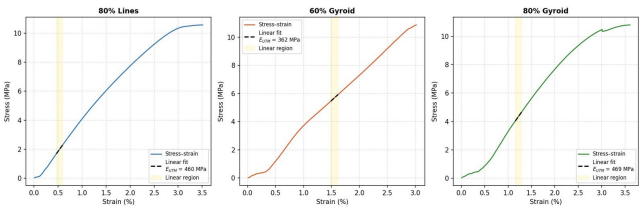

Fig. 10: Engineering stress–strain curves from UTM data for all three specimens. Dashed lines indicate linear regression fits within the identified Hookean window (shaded gold). E_{UTM} values from slope of each fit.

Table I summarises the modulus comparison across all specimens. In every case, $E_{\text{PINN}} > E_{\text{UTM}}$, consistent with the expected DIC-to-UTM scale effect: the IE-PINN recovers local (sub-gauge-length) constitutive stiffness from strain-gradient fields, while the UTM reports the specimen-averaged compliance that includes contributions from grip regions, free edges and infill porosity. The ratio $E_{\text{PINN}}/E_{\text{UTM}}$ ranges from approximately 1.4× to 1.9× across the three material architectures, suggesting that infill topology (Lines vs. Gyroid) modulates the scale-effect magnitude—a finding that motivates future multi-scale modelling.

The E_{UTM} values are consistently below E_{ref} for all specimens (by factors of 2.8–5.0×), indicating that the as-printed lattice specimens exhibit substantially lower macroscopic stiffness than their nominal design values—a known consequence of FDM-induced porosity, inter-layer adhesion defects, and the compliance of the infill junctions. The DHC-recovered E_{PINN} falls between E_{UTM} and E_{ref}, capturing an intermediate length-scale stiffness consistent with local deformation physics.

V. Conclusions

- Spatial reconstructions for all three material architectures—60% Gyroid ($t = 47.25$ s), 80% Lines ($t = 87.75$ s) and 80% Gyroid ($t = 111.0$ s)—maintain physical consistency under progressive loading. Critical metrics for the 80% Lines specimen ($v = -8.4$ mm, $\varepsilon_{xx} = 0.05$, $\varepsilon_{yy} = -0.168$) confirm robust field stability without oscillatory artifacts.
- UTM cross-validation across all specimens corroborates a systematic $E_{\text{PINN}} > E_{\text{UTM}}$ delta, with ratios ranging from 1.4× to 1.9×. The recovered local moduli ($E \approx 650$–912 MPa) versus compliance-averaged UTM baselines (362–469 MPa) validate the projected DIC-to-UTM scale-effect ratio.
- The DHC methodology accurately captures an intermediate-scale constitutive response, yielding E_{PINN} values positioned between macroscopic UTM results and nominal design moduli (E_{ref}), demonstrating high-fidelity stiffness identification beyond traditional testing capabilities.
- This mesh-free, data-efficient paradigm provides a scalable inverse mechanics architecture extensible to non-linear constitutive modeling, 3D structural configurations, and localized heterogeneous property mapping.

References

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